

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0501

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 35%

Code of Conduct:

I D. Gnana Deepthi (192411123)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

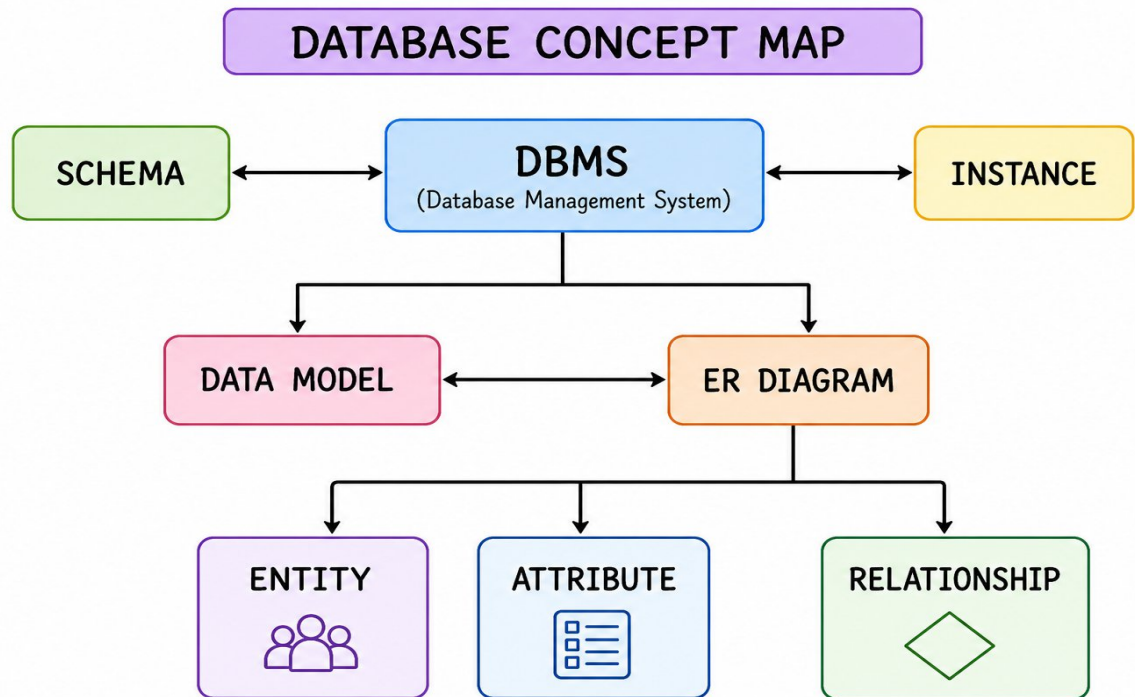
Signature of the student:

A rectangular box containing a handwritten signature in blue ink that reads "D. Gnanadeepthi".

QUESTIONS Marks: 50			Total
Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5
7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	5
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	5
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application	BL1 – Remember	5

	Developer.		
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1. Ans:



1. DBMS (Database Management System)

- DBMS is software used to create, store, retrieve, and manage databases.

2. Schema

- A schema is the blueprint or structure of a database.

3. Instance

- An instance is the actual data stored in the database at a particular time.
- It changes whenever data is added, updated, or deleted.

4. Data Model

- A data model defines how data is organized in a database.

5. ER Diagram (Entity-Relationship Diagram)

- An ER diagram is a graphical representation of the database design.
- It shows entities, attributes, and relationships.

6. Entity

- An entity is a real-world object or concept stored in the database.
- Each entity has its own attributes.
- Example: Student, Employee, Product.

7. Attribute

- An attribute describes the properties of an entity.
- Example: Student → Roll Number, Name, Age.

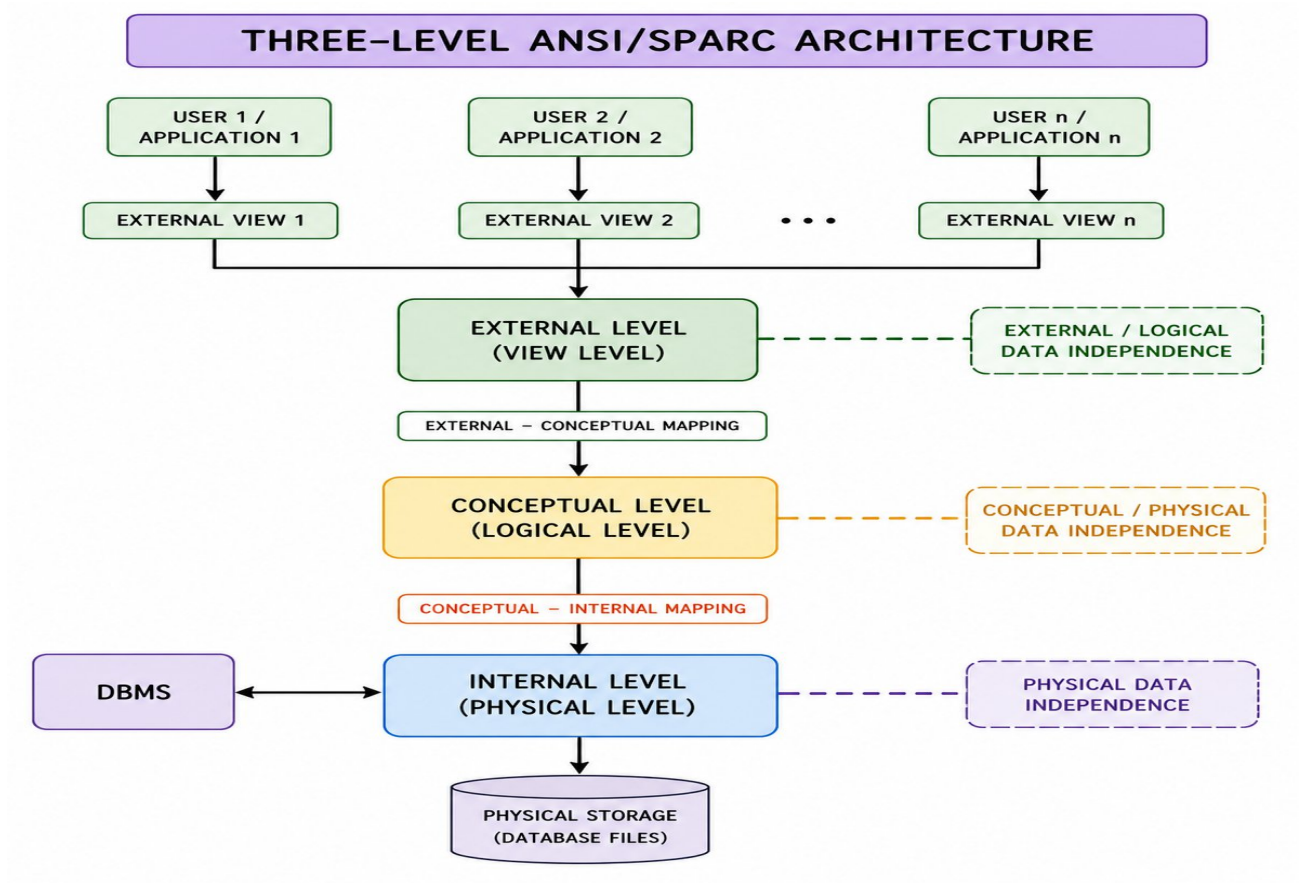
8. Relationship

- A relationship shows how two or more entities are connected.

Connection Between the Terms

- DBMS uses a Data Model to organize data.
- The Data Model is represented using an ER Diagram.
- The ER Diagram contains Entities, Attributes, and Relationships.
- The database structure is defined by the Schema.
- The actual data stored at any moment is called an Instance.

2. Ans:



1. External Level (View Level)

- Represents the user's view of the database.
- Different users can have different views.

2. Conceptual Level (Logical Level)

- Describes the complete logical structure of the database.
- Defines entities, relationships, attributes, and constraints.

3. Internal Level (Physical Level)

- Describes how data is physically stored.

4. External-Conceptual Mapping

- Connects external views with the conceptual schema.

5. Conceptual-Internal Mapping

- Connects the conceptual level with the internal level.
- Hides storage details from users.

Data Independence

Logical Data Independence

- Ability to change the conceptual schema without affecting external views.

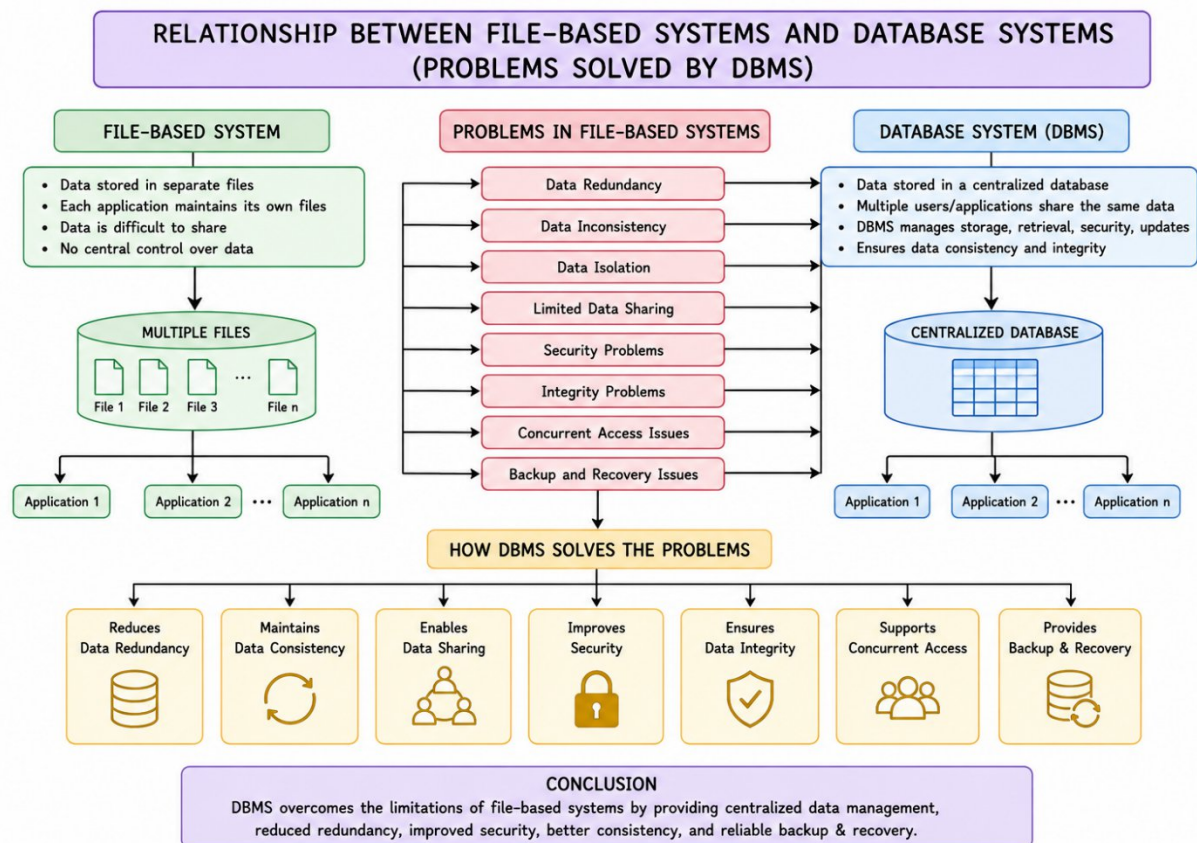
Physical Data Independence

- Ability to change the internal schema without affecting the conceptual schema.
- Example: Creating indexes or changing file organization for better performance.

Role of DBMS

- Manages all three levels of the architecture.
- Performs mapping between different levels.

3. Ans:



1. File-Based System

- Data is stored in separate files.
- Each application maintains its own files.
- Data is difficult to share between applications.
- No central control over data.

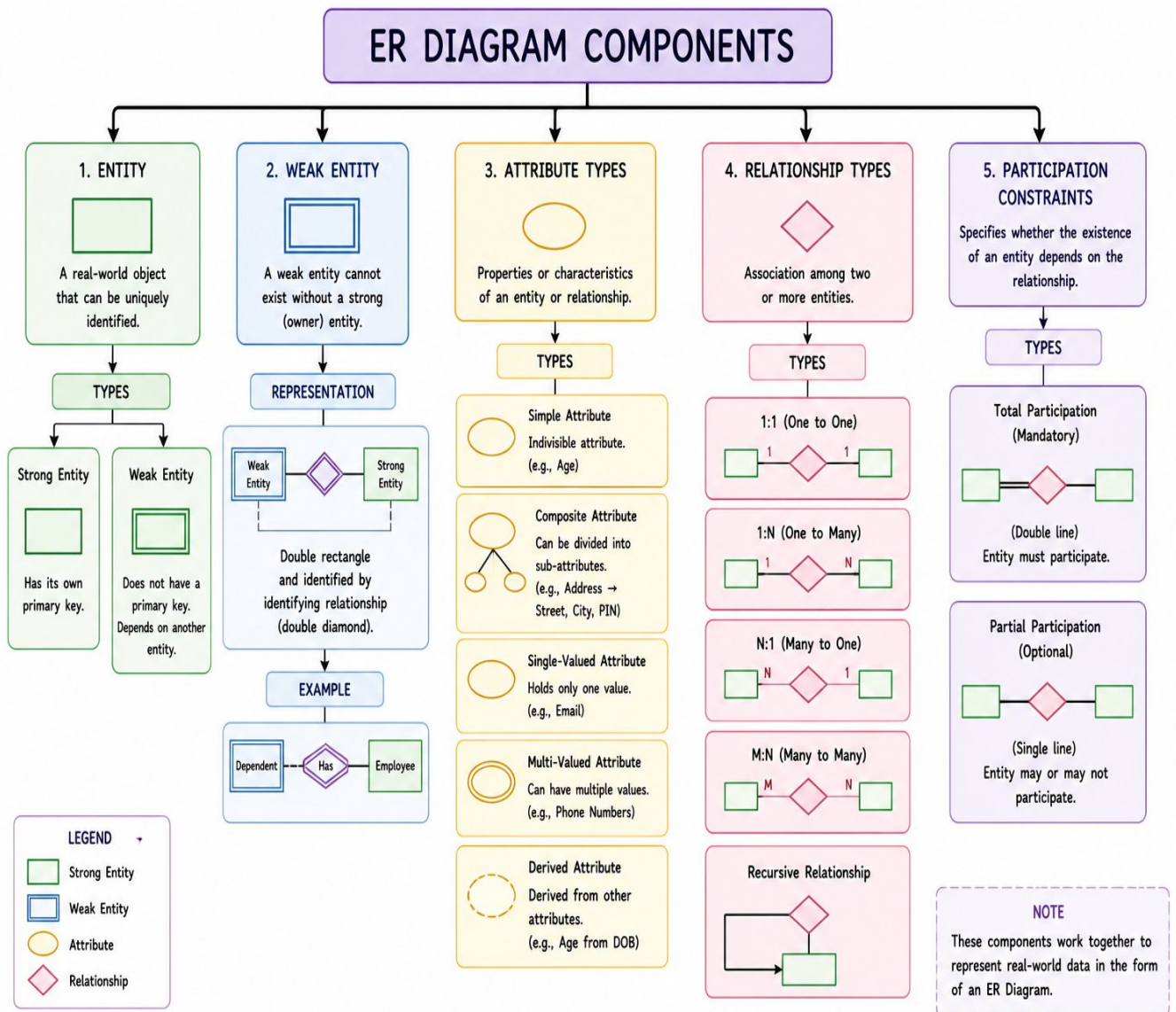
2. Database System (DBMS)

- Data is stored in a centralized database.
- Multiple users and applications can access the same data.
- DBMS manages storage, retrieval, security, and updates.
- Ensures data consistency and integrity.

Problems in File-Based Systems

- Data Redundancy
- Data Redundancy
- Data Inconsistency
- Data Isolation
- Limited Data Sharing
- Security Problems
- Integrity Problems
- Concurrent Access Issues
- Backup and Recovery Issues

4. Ans:



1. Entity

- A real-world object.
- Represented by a rectangle.
- *Example:* Student.

2. Weak Entity

- Depends on a strong entity.
- Represented by a double rectangle.
- *Example:* Dependent.

3. Attribute Types

- Simple: Cannot be divided (Age).

- Composite: Has sub-parts (Address).
- Single-valued: One value (Roll No.).
- Multi-valued: Multiple values (Phone No.).
- Derived: Calculated from another attribute (Age from DOB).

4. Relationship Types

- 1:1 – One to One.
- 1:N – One to Many.
- N:1 – Many to One.
- M:N – Many to Many.

5. Participation Constraints

- Total Participation: Every entity must participate.
- Partial Participation: Participation is optional.

5. Ans:

Hierarchical Data Model

- Organizes data in a tree structure.
- Each child has only one parent.

Network Data Model

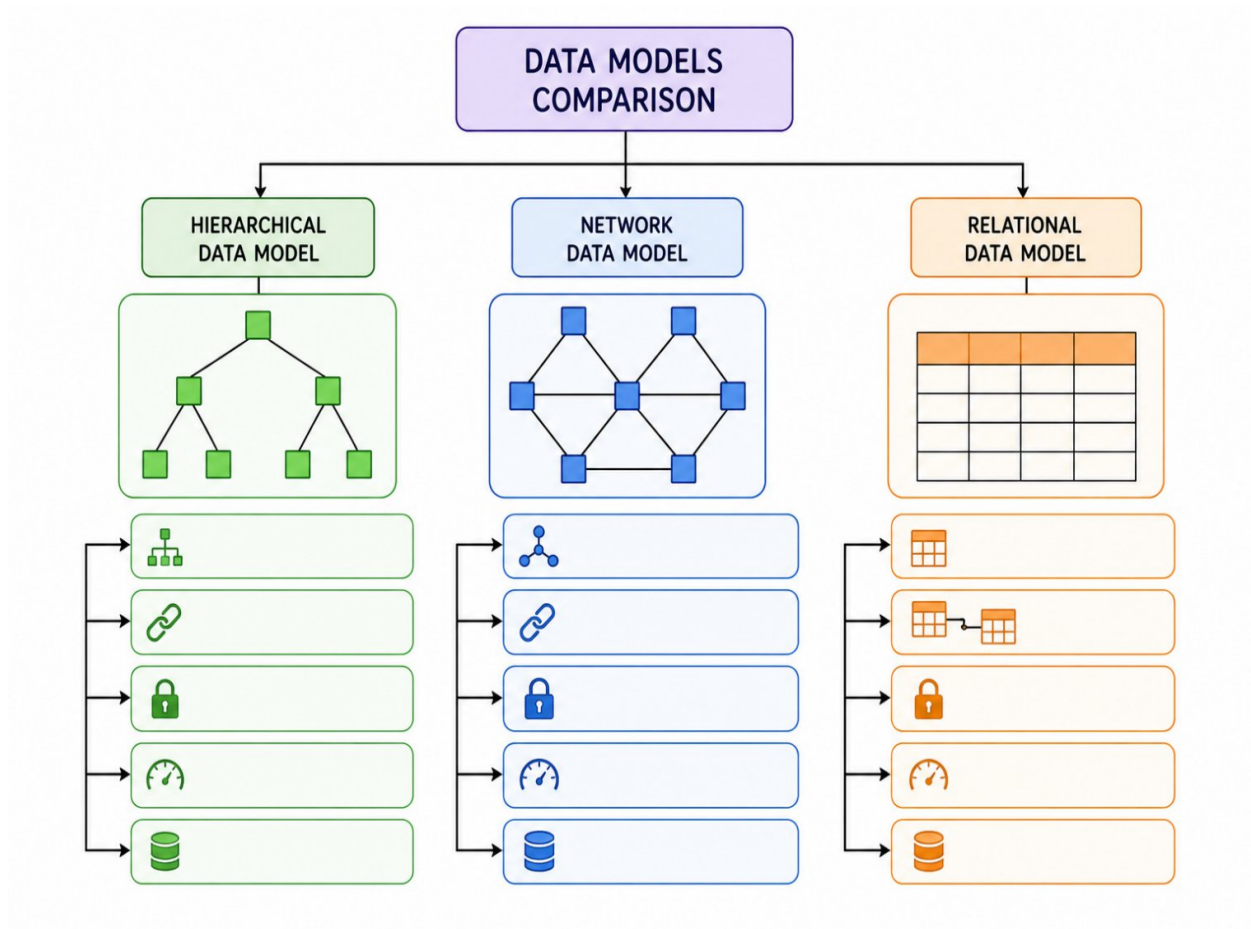
- Organizes data as a graph.
- A child can have multiple parents.
- Supports many-to-many (M:N) relationships.
- More flexible than the hierarchical model.

Relational Data Model

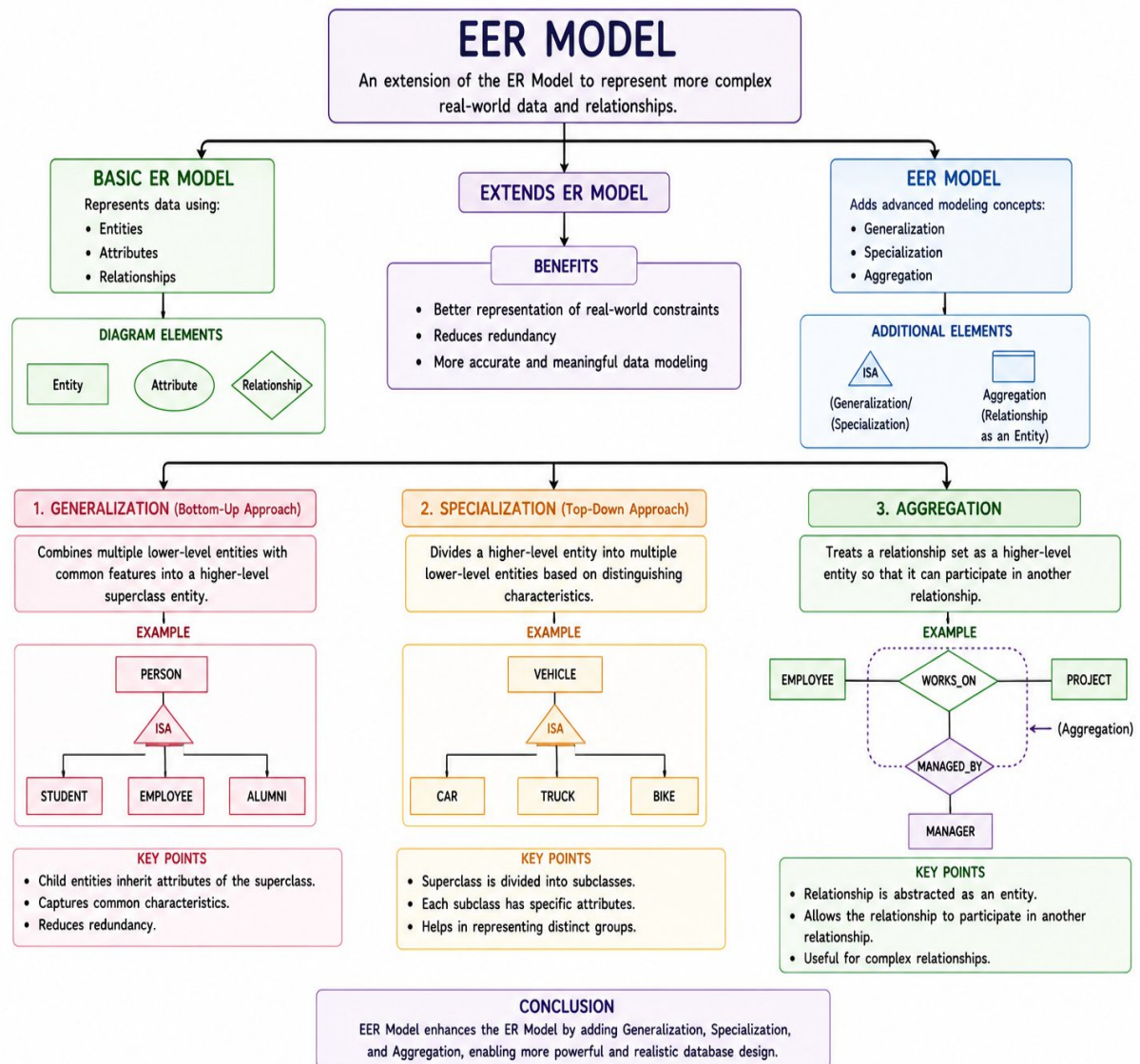
- Organizes data in tables (rows and columns).
- Tables are connected using primary keys and foreign keys.
- Easy to use and maintain.
- Most widely used database model.

Comparison

- Hierarchical → Tree structure, One Parent.
- Network → Graph structure, Multiple Parents.
- Relational → Tables, Key-based Relationships.



6. Ans:



7. Ans:

1. User / Application

- User submits a request (SQL query) to the DBMS.
- The request is sent through an application or interface.

2. Query Processor

- Checks the query for correctness.
- Optimizes the query for efficient execution.
- Creates an execution plan.

3. Storage Manager

- Manages data storage and retrieval.
- Handles security, transactions, and concurrency.
- Communicates with the database files.

4. Database Storage

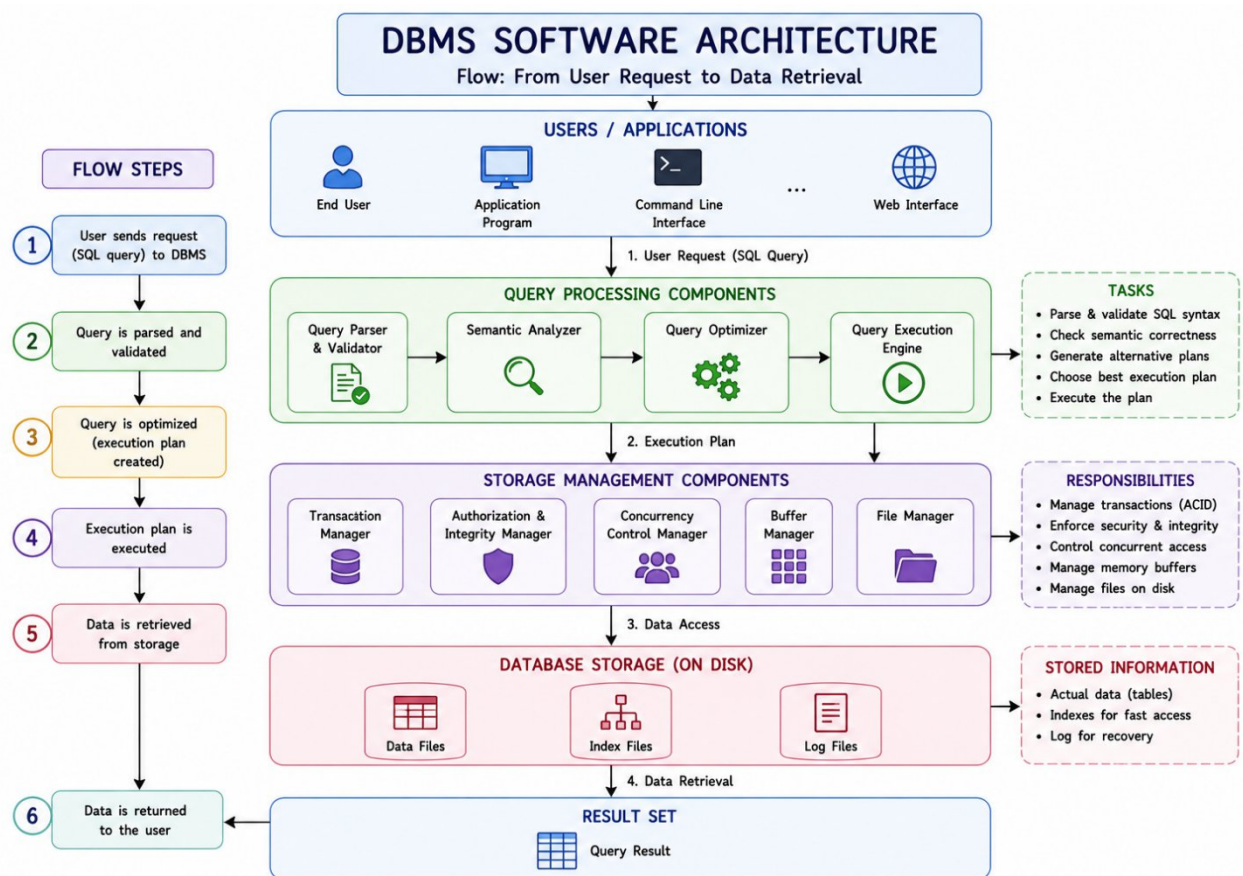
- Stores tables, indexes, and log files.
- Data is organized for fast access.

5. Data Retrieval

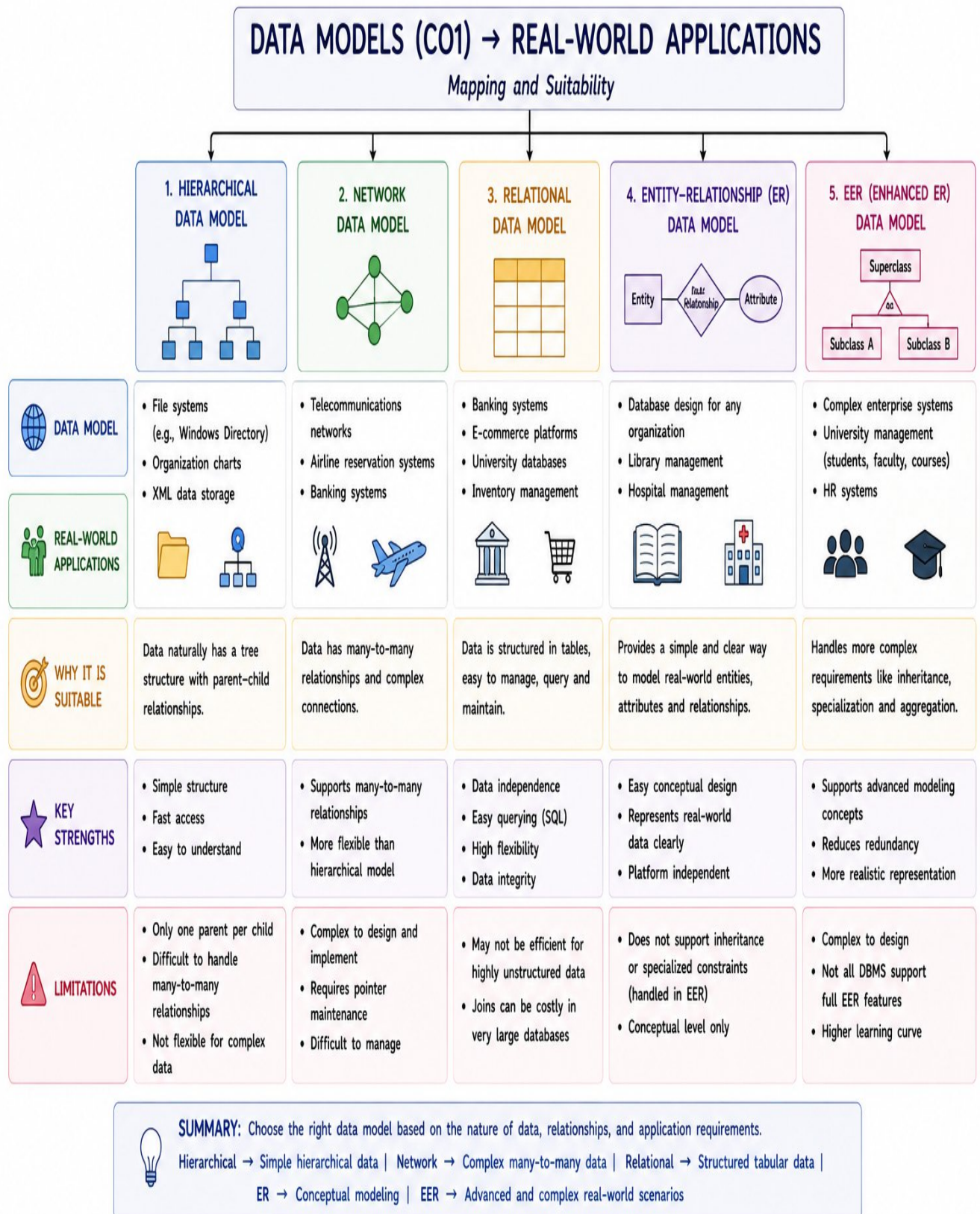
- The required data is fetched from storage.
- The DBMS processes the result.

6. Result Returned

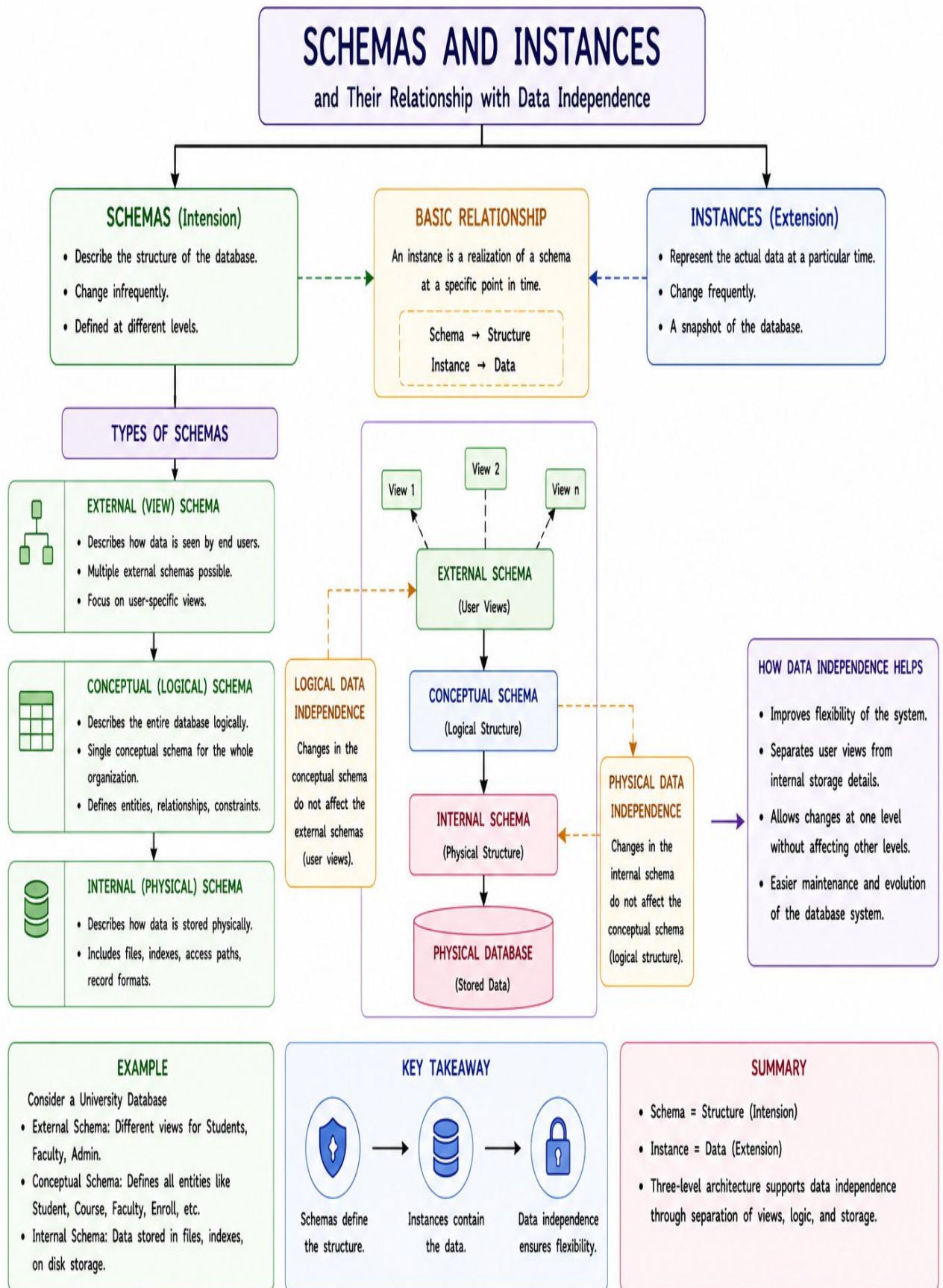
- The retrieved data is sent back to the user.
- Results are displayed through the application.



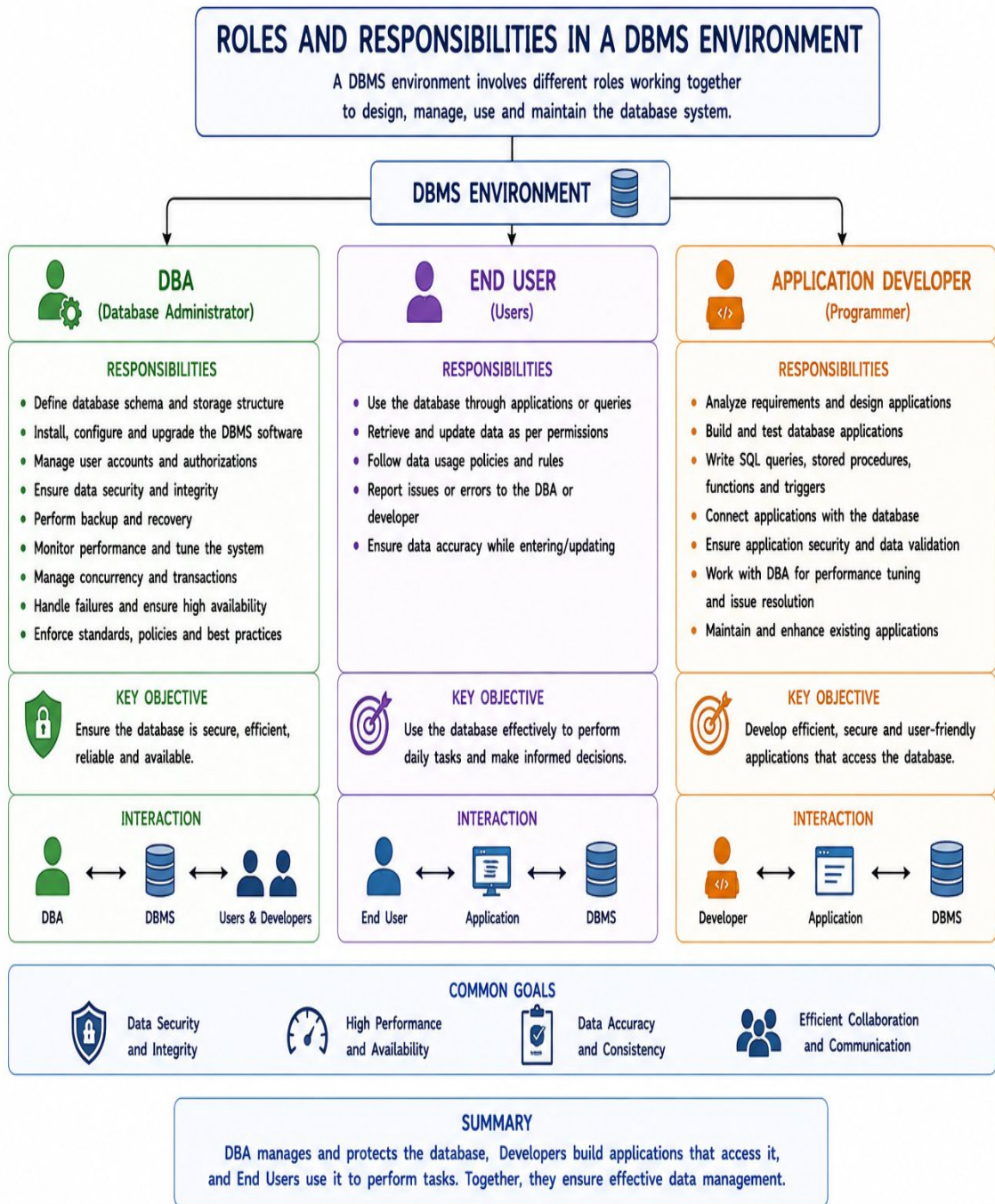
8. Ans:



9. Ans:



10. Ans:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

